

Antony Sequeira

Contact antony.sequeira@gmail.com +1 (510) 584-1426

Address 2659 Acorn St.
Castro Valley, CA - 94546

Legal status US Citizen

[LinkedIn](#) [Github](#) [Coursera](#)

Objective: Engineering role for design, development, and delivery of software systems.

Summary of experience

Over 20 years of experience in building backend systems from the ground up.

Worked in startups as well as large companies, and high growth rapid pace environments involving large volume applications. Proven track record of software design and development, taking products from concept to delivery.

Well versed in working out requirements with stakeholders, architecting the design and getting buy in, development, QA support, deployment, and operations support.

Skilled in implementing unit tests, stress testing, monitoring.

Consistently documenting requirements, design, and interfaces.

Work Accomplishments

[Xeduler](#) (Founder/Founding engineer) - Dec 2023 - to date

A customizable on demand workers scheduling system for small businesses.

This system was developed and is run by me. The backend service and jobs (Python) are deployed ([CDK-tf](#)/Python) to GCP (CloudSQL/CloudRun). Developed and deployed web, Android, iOS ([Flutter](#)) apps for frontend.

Contract work (Software Consultant/Engineer) Nov 2017 - to date

- Produced a self contained AWS Lambda/Python/Terraform based system for providing a PII redaction system across multiple document types - worked out requirements, proposed architecture, implemented, and deployed. This proved critical for the startup to produce

their MVP and get VC funding.

- Enhanced the redaction system to support images using aws tesseract and improve the semantics using Google DLP.
- Honeywell thermostat cloud data collector (Java/Storm) - EcoFactor contract.
- Personal project (python/postgres backend with Angular 5 frontend with AWS setup).

Polysign - Sr. Software Engineer (remote) Aug 2021 - Nov 2023

- Design and development work on the distributed trading product backend (k8s/Python/Postgres).
- POC for an alternative to the company's existing identity/auth platform (keycloak)
- Improved product quality by adding proper transaction envelopes for the API and messages processing.
- Improved product quality using API performance testing (introduced the *Locust* framework to the company).

Granular San Francisco, CA - Sr. Software Engineer Oct 2016 - Oct 2017, July-2018 - July-2021

- Designed and built an automated system for using satellite tasking api to go from a geo shape to downloading the final images (Python AWS Lambda and DynamoDB)
- Microservice infrastructure for hosting deep learning models (Python, Tensorflow, Kubernetes/Docker, Postgres, Terraform, AWS).
- Design and implementation of mobile sync backend prototype.
- Designed and implemented the mobile activity feed backend (Python / SQLAlchemy / Postgres) .
- Farming equipment operator CRUD features.

StumbleUpon San Francisco, CA - Sr. Software Engineer Mar 2015 - Sept 2015

- Designed and developed mobile push for daily digest (Scala/Akka/Reactive/RabbitMQ).
- Deployment automation (ansible).
- Maintain and enhance legacy Php code.

EcoFactor Inc., Redwood City, CA - Sr. Software Engineer Sep 2012 - Feb 2015

Energy savings and demand response through residential thermostats management. I was a major contributor to the platform engineering team as we scaled over the last two years:

- Designed and built the new data collection platform using twitter storm (base framework, centralized logging, integration with kafka, Javascript based configuration, health server, and the data collection application) in Java.
- Designed and built an ETL infrastructure for time-series data (Python/Hive-Hadoop/Sqoop). Works with real and synthetic time-series data with ability to add more sources via declarative logic. Designed to work with near zero sysops overhead.
- Built a server to simulate partner data feed. This started the trend of using mock servers (Node.js/Javascript) within the company for functional and stress testing purposes.
- Worked with devops to get centralized logging (logstash) and automated deployments

working for the platform team.

- Used the ETL based mechanism to also power the recommendation application so that it can scale better (python, hive streaming)
- Provided a publishing mechanism to match arbitrary data objects to mysql tables for publishing results from hadoop/hdfs/hive to mysql (python)
- Took over existing data processing cron jobs. Made them reliable and not require babysitting (automated progress tracking, simplified recovery and rebuilds - shell, Java)
- Implemented Java data apis as needed (legacy Java/tomcat)

SunPower, San Jose, CA - Sr. Software Engineer Sep 2008 - Jan 2012

Played a crucial role over the initial and subsequent releases of the residential customer applications as the core developer of the API server:

- Took over maintenance of existing customer and admin web UIs (C#/ASP.NET)
- Designed, developed (C# .net) and maintained the api server over multiple releases serving iPhone, Adobe Flex, and Android UIs. The API server provided authentication and data services for the SunPower residential customers' data. This involved designing and developing the overall framework, web security (multi domain single sign on) and all the layers.
- Designed and implemented the first SunPower Android App (Java/Android)
- Helped provide clarity on requirements to DB and UI (iPhone, Flex) engineers

I worked for SunPower as independent contractor after Sep 2010

AdBrite Inc., San Francisco, CA - Software Engineer Feb 2008 - Sep 2008

A high volume ad network delivering more than 750 million hits per day. In the short time I have worked here, I quickly came up to speed and delivered the following features on the ad server (Php/Linux):

- Integrating new search backend (rpm, ant, http client fixes)
- Helped in the move from MySQL to fully memcached backed ad serving
- Behavioral targeting component for ad server, host based ad matching using third party categorization, URL based ad matching using third party categorization
- Integrating age and gender targeting

Simply Hired Inc., Mountain View, CA - Founding Lead Software Engineer Dec 2004 - Jan 2008

This is a vertical search startup for aggregated job advertisements. As part of the initial team of four engineers, I was instrumental in many areas in getting the initial product out (March 2005). As a result the company was able to raise 20 MM in under 2 years.

- Played a critical role in going from no product to live beta in less than three months.
- Developed (as the initial sole UI engineer) the main product UI www.simplyhired.com server (PHP/Mod-Perl/Apache/MySQL/Javascript/XHTML).
- Worked as the lead search engineer for the development of the search back end services (Java/Servlets/Lucene).
- Provided engineering clarity to the 'sponsored job listing' initiative and implemented the

indexing and searching backend for it (Java/Lucene).

- Developed 'location' processing of job data (Perl) and designed a simple algorithm for the distance search.
- Designed and developed a web crawler prototype including a high performance NIO based fetcher (Java).
- Designed and implemented a unique, simple technique to measure any web application's maximum capacity for requests per second.
- Designed and developed a rule based job ranking service with a Javascript(Rhino) based rule engine (Java/Jetty/Rhino/MySQL) with capability to plug in more engines.
- Came up with a solution for implementing high performance collapsing of results on seemingly identical results.
- Spearheaded Wiki usage.
- Designed and implemented initial production deployment mechanisms.
- Designed and implemented a simple algorithm for job de-duping (Perl).

AOL / Netscape / AtWeb, Sunnyvale, CA - Sr. Software Engineer Dec 98 - Dec. 2004

Remained productive and delivered consistently through all of the mergers and changes in directions and technologies.

- Performance measurements and improvements of the very high volume AOL Polls product (Java/Resin/Solaris) during the peak load of the last Presidential election.
- Design and development of components of the AOL blogs product (Java/Struts/Tomcat/Linux) and design of the URL structure.
- Design and development of AOL message boards web front-end servers (C/C++/Tcl/AOLServer/Linux/Solaris).
- Search sub-systems integration for AOL message boards backend servers (C/C++/Tcl/Solaris/Linux).
- Development of news feeder for AOL Alerts (C/Solaris).
- Maintenance and development of new features of 'Register-It' product (Perl/C/C++/Linux).

Siebel / Scopus, Emeryville, CA - Lead Engineer Nov. 97 - Nov 98

- Integration of MS VBA into Siebel tools and client (Visual C++/Win32).
- Developed a UML model for a new SCOPUS architecture prototype.
- Implement integration facility for OCX controls and Java Beans into the SCOPUS product (Visual C++/ Java JDK 1.x).
- Internationalization of the look and feel of the SCOPUS product.

EC Company, Palo Alto, CA - Sr. Software Engineer Jan. 97 - Oct. 97

- Enhancing 'EC Exchange' (Visual C++ /MFC).
- Communication server (Visual C++ /MFC).
- InstallShield installer creation for the 'EC Exchange' client.
- SEF (EDI metadata) parser (Java/JDK1.1)
- Online client enrollment application prototype, generic wizard sheet, Javabeans for numeric input with a built in calculator, JDBC wrapper classes and mutable data type classes for data

binding (Java).

Synectics, Bangalore, India - Founding Engineer/Engineering Manager May 95 - Dec 96

I was the technical architect for the company, taking the basic idea of a retail banking system to an implementation that is ready for customization for their first customer. The system was designed to be distributable across unreliably connected bank branches (third world market). It was highly customizable and object oriented from the ground-up.

Technical accomplishments as the founding engineer/architect:

- Designed the overall architecture of QUAESTOR suite of products for bank branch automation.
- Detailed design of the retail-banking product.
- Business functionality and user interface development.
- Interaction with other products in the suite.
- Design of the class hierarchies, individual classes and their inter relations.
- Configuration for user security, access control, product customization.
- Development of general-purpose component class libraries (Win32, Visual C++, MFC, ODBC).
- Design of the database tables (MS Access).
- Development of the core business classes and the application framework(C++).
- Development of OCX (ActiveX) controls.
- Development of the framework for the message queuing server.
- Customizing the product and upsizing to SQL Server 6.5 for the first client in London, U.K.

Management accomplishments:

Managing and mentoring four software engineers, providing technical leadership to the other technical managers in developing the complete products suite.

- Facilitating teamwork and scheduling tasks (Visual Source Safe and MS Project).
- Interacting with the marketing team.
- Documenting the functional and design specifications.
- Performance review of programmers.

United Saudi Commercial Bank, Riyadh, Saudi Arabia - Programmer Analyst Mar 92 - Apr 95

All of the applications here were bilingual (Arabic and English) and worked 'Right to Left' when in Arabic mode.

ESISNET – Stock Trading Terminal

- Designed the system and developed an interpreter (C/DOS) for a custom language.
- Application workflow and communication with OS/2 server (AAI, SNA, X.25) using the above interpreter defined language. Internationalization and character-set translation (Nafitha to IBM NLS).

ESISLINE – Stock market feed gateway

Multithreaded multi serial port file transfer from OS/2 to AS/400 - application design and development (C OS/2)

Retail Banking System

- Development of the functional and design specifications.
- Client-server application design and development (Win 3.1 Visual C++ 1.52, MFC).
- Database design and stored procedure coding. (Transact SQL).
- AS/400 application interface design and development (SNA Server, NetSoft API, 5250 Emulation).
- Reusable text editor component for use for editing and report viewing (C/DOS)

Middle East Traders, Doha, Qatar - Programmer and System Administrator Mar 90 - Mar 92

I was **solely** responsible for the information system of the company, which was mission critical for the day-to-day operations.

- Software (inventory, accounting, payroll, workshop) maintenance, and development of new features (COBOL).
- System administration (NCR 32/450, UNIX System V) and user support.

Indian Space Research Organization, Bangalore, India - Engineer Feb. 88 - Mar 90

Developed all of the software needed for running a single board computer (8086 SBC with Synchronous and asynchronous serial ports, timer, interrupt controller etc.), including boot-up, dumb-terminal management, multitasking control, and the applications. This system provided real-time acquisition and analysis of satellite housekeeping telemetry and a 'Tele-command Console' (Embedded software PL/M86).

BPL, Bangalore, India - Trainee Engineer Jul 87 - Jan 88

- Troubleshoot and fix faulty defibrillators and ECG monitors on the production line.
- Designed and developed a digital post alarm ECG printer.

Education

Master of Science - Data Science - University of Colorado, Boulder - 2021 - In progress.

[Coursera](#) Machine Learning Specialization courses.

Continued education courses from *Indian Institute of Science* (1989-90), Bangalore, India.

Bachelor Of Engineering (1983 - July 1987) - Electronics and Communications - K.R.E.C, Surathkal, Mangalore University, India (presently known as *National Institute of Technology, Karnataka*).